Longitudinal dynamics - Dynamic Force analysis of a Vehicle in Acceleration – exercise

If a Front wheel drive car is generating a traction force (at wheel) of 2000 N at a given torque/ speed
The car is travelling at 30 kmph steadily and then accelerates by introduction of torque at wheel
What is acceleration on flat ground ?

Vehicle details
Weight = 1500 kg
CG height = 350 mm
Wheelbase = 2300 mm
CG location 1000 rear of front
Rolling resistance coefficient = 0.0145
Frontal cross-sectional area * drag coefficient = 0.50 m^2
Height of effective Drag force from ground = 300 mm

$$\boldsymbol{F_{xf} + F_{xr} = \left(+ \left(\frac{W}{g}\right)(a_x)\right) + R_{xf} + R_{xr} + W sin\theta + D}$$

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$$W_f = \frac{-Dh_A - \left(\frac{W}{g}a_x h\right) - W sin\theta h + W cos\theta b}{L}$$

$$W_f = \frac{-Dh_A - \left(\frac{W}{g}a_x h\right) + Wb}{L}$$

$$W_r = \frac{+Dh_A + \left(\frac{W}{g}a_x h\right) + W sin\theta h + W cos\theta a}{L}$$

$$W_r = \frac{+Dh_A + \left(\frac{W}{g}a_x h\right) + Wa}{L}$$